CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 6: 0/1 Knapsack Problem with Dynamic Programming

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #6 | Academic Year: 2026-27

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1. Overview & Objective:

Construct the DP bottom-up tabular state matrix and backtrack to identify selected items.

2. Practical Tasks & Verification Steps:

1. Construct 2D integer array dp[n+1][W+1] where n is number of items and W is knapsack capacity.
2. Populate table using recurrence dp[i][w] = max dp[i-1][w], val[i-1] + dp[i-1][w - wt[i-1]] .
3. Backtrack from dp[n][W] to identify exact set of selected items contributing to optimal value.
4. Verify time complexity of O nW and compare efficiency against recursive exponential brute-force.

Submission Guidelines & Verification:

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